Feature-Based Global Wheat Full Semantic Segmentation using Classical Machine Learning and Contour Analysis

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***Abstract*—Wheat is considered as one of the essential crop in terms of global food security. Accurate crop monitoring is important to improve agricultural productivity and supporting sustainable farming. Nowadays, deep learning is widely used for automated crop analysis, but these models usually act like "black boxes" and are difficult to interpret. Our project presents a classical machine learning-based framework for wheat-head semantic segmentation using the Global Wheat Head Dataset 2021 (GWHD2021). Instead of depending on complex black-box models, our approach focuses on physical characteristics of wheat, such as texture, shape,and colour. By using hand-crafted features, the system becomes more transparent, easier to understand, and more reliable for agricultural researchers. Our work provides a basis for accurate wheat-head segmentation and sets the stage for future boundary refinement. Overall, our work provides a practical and easy-to-understand solution for plant phenotyping, crop monitoring, and agricultural image analysis.**

***Index Terms*—GWHD 2021, Classical Machine Learning, Wheat Phenotyping, Semantic Segmentation, Hand-crafted Features**

I. Introduction

Wheat is one of the essential food crops worldwide. This makes precise monitoring of the crop essential to improve productivity and resource use. Automated plant phenotyping use to analyze key parts of plants, such as wheat heads, leaves, and stems, for a better understanding of crop growth and health. The Global Wheat Head Dataset 2021 (GWHD2021) supports this by providing diverse high-resolution images from different environments and growth stages.

Although deep learning is widely used in agricultural computer vision, these models often work like "black boxes" and are difficult to interpret. To overcome this, our project uses a classical machine learning approach that focuses on clear and explainable features such as shape, colour, and texture. This makes the segmentation process easier to understand and provides a more practical and trustworthy solution for researchers.

II. Methodology

A. Feature Engineering Pipeline

The results of machine learning semantic segmentation model depend on the quality of its handcrafted features. Because non-neural classifiers like Random Forests cannot automatically identify hierarchies or visual patterns from raw image matrices, we have to create a feature space that clearly describes the characteristics of each pixel.

To accurately classify pixels into wheat spikes, stems, leaves or background our feature engineering pipeline extracts three core categories of descriptors: Color, Texture, and Shape.

B. Color Descriptors

We start with RGB images, but they are sensitive to variations in field description, camera exposure, and shadows, which can severely degrade pixel-level classification. To achieve invariance, we convert the RGB images into the CIELAB color space. The L-channel represents grayscale intensity, isolating the geometry and shading of the wheat plant as the main input for subsequent texture and shape filters. The a-channel and b-channel represent the red and blue-yellow spectral regions. These raw chromaticity values are extracted directly as color features, so the model can recognize a leaf in bright sunlight or a dark shadow.

C. Texture Descriptors

Color alone is not enough to distinguish between plant organs of hues such as a yellow leaf and a yellow wheat spike. To solve this problem, we extract structural patterns using two distinct texture analysis methods applied to the L-channel:

- Local Binary Patterns: This acts as a texture analyzer. It compares the intensity of a pixel to its surrounding neighborhood and generates a binary code. This feature identifies the grainy and bumpy surfaces characteristic of dense wheat spikes. it mathematically distinguishes them from the smoother surfaces of leaves and stems.
- Gabor Filter Banks: To capture macro-textures and directional orientations we apply Gabor filters. These linear filters analyze frequencies at specific angles. Because wheat plant architecture features extended stems, applying directional Gabor filters enables the model to detect these linear patterns, making it a highly effective stem detector.

D. Shape Descriptors

- In traditional classification, true geometric shapes cannot be calculated for an independent pixel. To overcome that,

we use edge detection as a proxy for structural boundaries. We apply the Sobel Gradient Magnitude operator to the L-channel to compute the first-order derivatives of pixel intensities. The Sobel filter effectively highlights regions of spatial frequency, outlining the wheat components. By mapping structural changes, the model is provided with the precise boundaries where a stem overlaps a leaf or where a wheat head stands against the soil background.

E. Feature Aggregation

- Once the individual feature maps are computed the 2D arrays are flattened into vectors and concatenated. This transforms the image data into a high-dimensional tabular dataset, where every pixel is represented as an independent row of engineered features ready to be ingested by the classical machine learning classifier.

F. Creating Training Data (Otsu's Thresholding)

- To train the machine learning model without spending too much time on manual annotations we created fake training data using Otsu's Thresholding. This method helps find the threshold to separate wheat spikes from the soil and foliage by making the differences within each group as small as possible. The resulting black and white images serve as the target data for each pixel during training with Otsu's Thresholding.

G. Training the Model (Random Forest)

We then use this dataset to train a Random Forest Classifier. We chose this model because it is very efficient when dealing with millions of pixel data points. By looking at the color, texture and shape of the wheat the Random Forest learns to classify wheat components that're close together. The Random Forest model effectively handles the data required to classify dense wheat spikes.

III. Results

A. Exploratory Data Analysis

1) Color Space Separability: Color space plots demonstrate overlap between spike and background; hence, we need to use textural descriptors for accurate pixel-level classification

2) Texture Discriptor Analysis: Comparison of grayscale patches and their corresponding Local Binary Pattern textures. It gets the complex geometry of wheat kernels, transforming into a mathematically distinct textural feature.

3) Edge Density and Gradient Magnitude: Analysis of image gradients for the initiation of Active Contour. The Sobel filter (center) shows the spatial gradient magnitude, which serves as the energy map used to change the contour shape. The Canny edge detector (right, $\sigma = 1.5$) shows the clear morphological differences between the high-frequency internal texture of the wheat spike and the continuous, low-frequency edges of the background foliage.

4) Pixel Level Class Distribution: Shows the Pixel-level distribution. Class imbalance in the dataset: 3:1 ratio of background pixels to spike pixels. We'd require a balanced training set for Random Forest or SVM classifiers.

B. Model Performance and Interpretability

1) Quantitative Performance & Feature Importance: There's class imbalance, so the Random Forest was trained on a balanced dataset, achieving an Intersection over Union of 0.2371 on the test set. The model relied most on L-channel and vertical/horizontal Gabor features.

C. Visual Validation and Error Analysis

1) Qualitative Assessment & Limitations: The model can localize wheat heads, but it also segments leaves and stems, when we are trying to do only wheat head segmentation. It relies heavily on Lightness, so it produces false positives on leaves and stems under direct sunlight. Additionally, our pseudo-ground truth, using Otsu thresholding, labelled trapped background soil and shadows as wheat spikes, which adds noise in the training data.

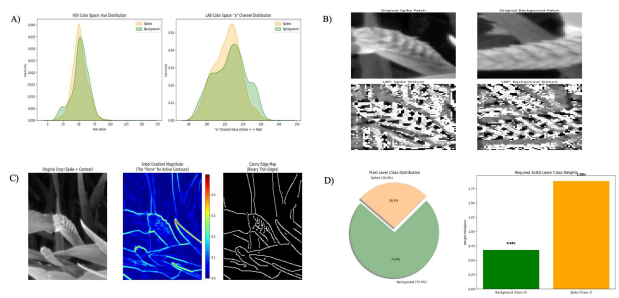

Fig. 1. EDA applied on GWHD2021 : A) Color Space Separability, B) Texture Discriptor Analysis, C) Edge Density and Gradient Magnitude, and D) Pixel Level Class Distribution

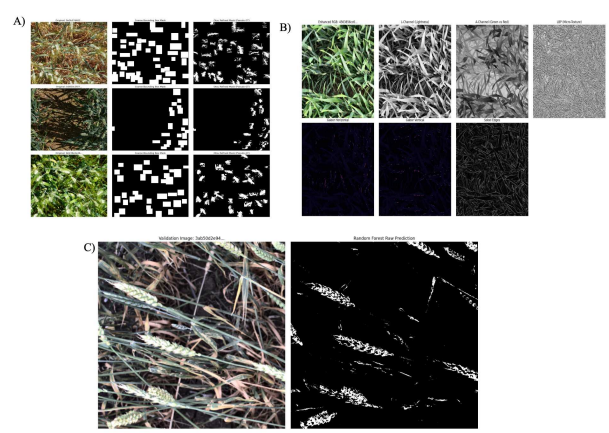

Fig. 2. Feature Extraction and Segmentation Stage : A) Pseudo-Ground Truth generation, B) Selection of hand-crafted color, textural and structural feature maps, and C) Semantic segmentation output from the trained Random Forest classifier

IV. Conclusion

In this work, we developed a feature based approach to analyze wheat images using classical Machine learning techniques. Peformed different types of features such as color information from the LAB color space and texture related features like LBP and Gabor were extracted to represent important patterns in wheat images. The bounding box annotations from the csv file were processed to create binary masks so that model could learn from labelled training data properly. After that used Random Forest to identify wheat heads based on these features. So the results showed that traditional image processing and machine learning methods can be useful for analyzing crop images

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